

26) Both A and B say "I am a Knight"

Soln: A Knight, who always speaks the truth can say "I am a Knight" because it is the truth. A Knave, who always lies can say "I am a Knight" because it is a lie. $\therefore$ The following 4 situations arise:

① $A \rightarrow \text{Knight}$ $B \rightarrow \text{Knight}$ ② $A \rightarrow \text{Knight}$ $B \rightarrow \text{Knaves}$ ③ $A \rightarrow \text{Knaves}$ $B \rightarrow \text{Knight}$ ④ $A \rightarrow \text{Knaves}$ $B \rightarrow \text{Knaves}$

27) A says "We are both Knaves" and B says nothing.

Soln: If A is a Knight, "We are both Knaves" must be True. But A is a Knight. This is a contradiction.

$\therefore$ A must be a Knave. So, the statement "We are both Knaves" must be False. $\therefore$ B can't be a Knave and hence must be a Knight.

$A \rightarrow \text{Knave}$ $B \rightarrow \text{Knight}$

Exercises 28-35 relate to inhabitants of an island on which there are three kinds of people: Knights who always tell the truth, Knaves who always lie, and spies (called normals by Smullyan) who can either lie or tell the truth. You encounter three people, A, B and C. You know one of these people is a Knight, one is a Knave, and one is a spy. Each of the three people knows the type of person each of the other two is. For each of these situations, if possible, determine whether there is a unique solution and determine who the Knave, Knight and spy are. When there is no unique solution, list all possible solutions or state that there are no solutions.

28) A says "C is the Knave," and B says "A is the Knight," and C says "I am the spy."

Soln: Let A be the Knight. $\therefore$ C is the Knave, $\therefore$ a Knight always tells the truth. C says "I am the spy" which is consistent with it being a Knave, since it is a lie. $\therefore$ B must be a spy, and a spy can speak the truth sometimes. There is no issue with B being a spy since "A is the Knight" is True.